

Time and planning observations

Connect the clock to the sky

WVU DSPIRA astronomy

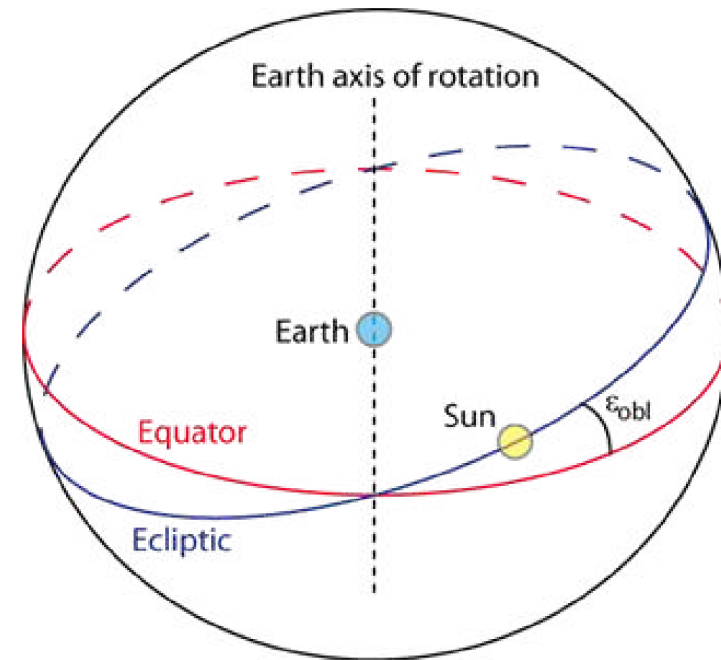
Adapted from lecture material credited to Loren Anderson

Two reference planes in the sky

The celestial equator extends Earth's equator onto the sky.

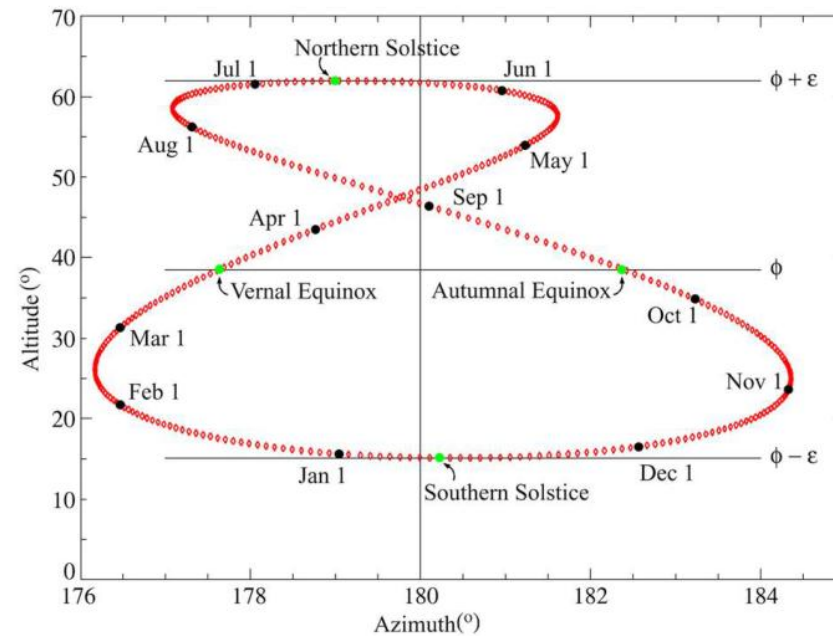
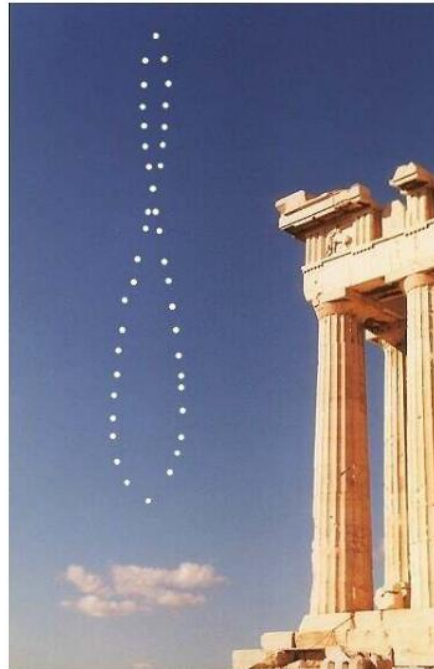
The ecliptic traces the Sun's apparent annual path.

Earth's tilted axis separates these planes.



Earth's equator and the ecliptic define different planes.

The analemma



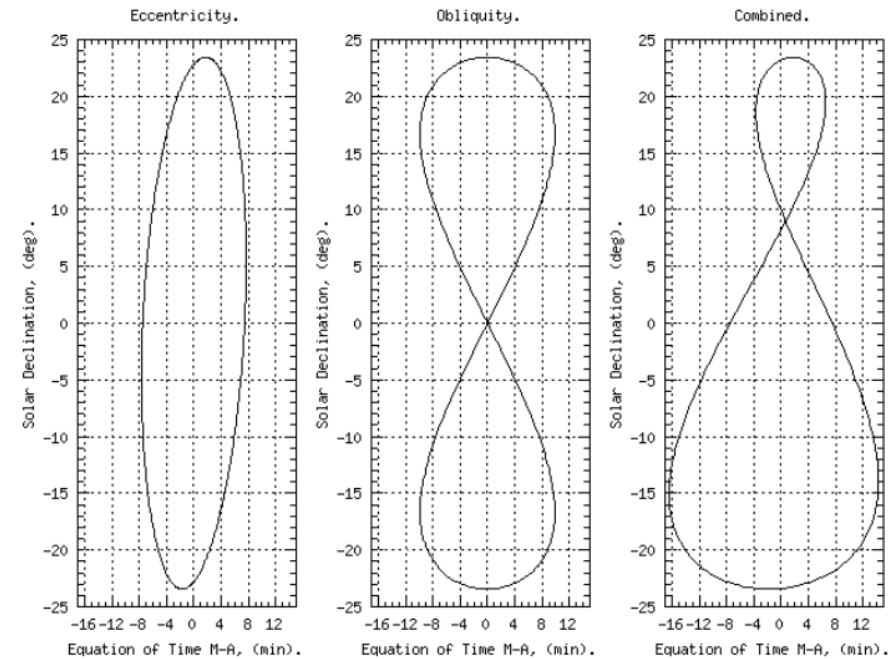
Record the Sun from one location at the same mean solar time throughout a year.

Why the analemma forms

Earth's orbital speed varies.

Axial tilt changes the projection of solar motion.

Both effects change apparent solar time.



Orbital eccentricity and axial tilt combine to produce the analemma.

The equation of time

Apparent solar time

Tracks the real Sun.

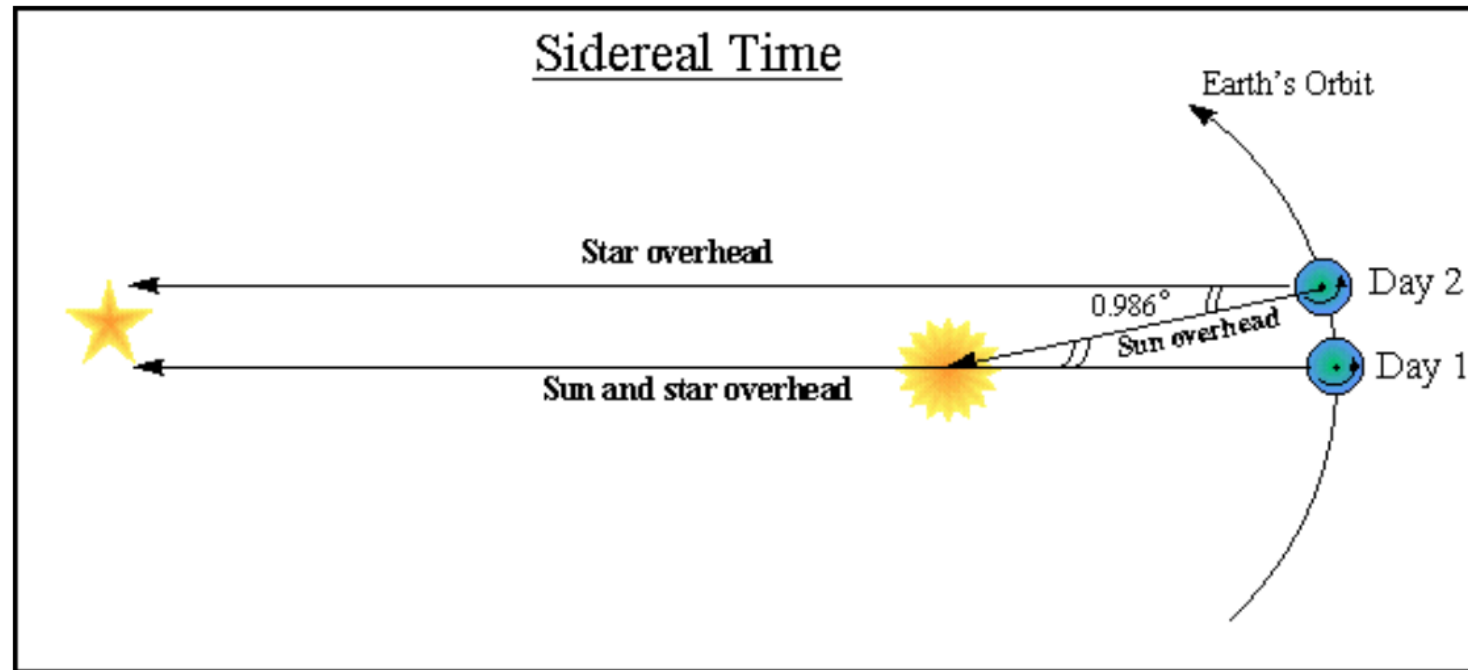
Apparent noon occurs when the Sun crosses the local meridian.

Mean solar time

Uses a uniform solar clock.

Equation of time =
apparent solar time
– mean solar time

A sidereal day follows the stars



A sidereal day is about 23 hours 56 minutes. A mean solar day is 24 hours.

Local sidereal time

Read the sky clock

Local sidereal time is the right ascension crossing the upper meridian.

It depends on both time and longitude.

Use hour angle

$$HA = LST - RA$$

At upper transit, $HA = 0$.

An object transits when LST equals its RA.

Practice with hour angle

Local sidereal time	Target RA	Hour angle
4 hours	6 hours	-2 hours: east of meridian
6 hours	6 hours	0 hours: upper transit
8 hours	6 hours	+2 hours: west of meridian

Hour angle increases as Earth rotates. Upper transit is usually a useful time to observe a visible target.

Compare the equinoxes

March equinox

The Sun's RA is approximately 0 hours.

At local apparent noon, LST is approximately 0 hours.

September equinox

The Sun's RA is approximately 12 hours.

At local apparent noon, LST is approximately 12 hours.

Calculate local sidereal time

1. Enter the observing date and UT1 time.
2. Enter longitude using the calculator's sign convention.
3. Find LST and compare it with your target's RA.
4. Check altitude and obstructions in Stellarium.

[Open the U.S. Naval Observatory sidereal-time calculator](#)

Plan an observation

Choose the target and record its coordinates.

Find a time when it is above your local horizon.

Record the site, date, time zone, azimuth, and elevation.

Check that your clock settings match your observing plan.

Reference tools

[U.S. Naval Observatory sidereal-time calculator](#)

[U.S. Naval Observatory equation of time guide](#)

[Stellarium user guide](#)

Original lecture: Loren Anderson